

Session 12 (AIML) – Assignment Question

Name of Student : Ziya Shaikh

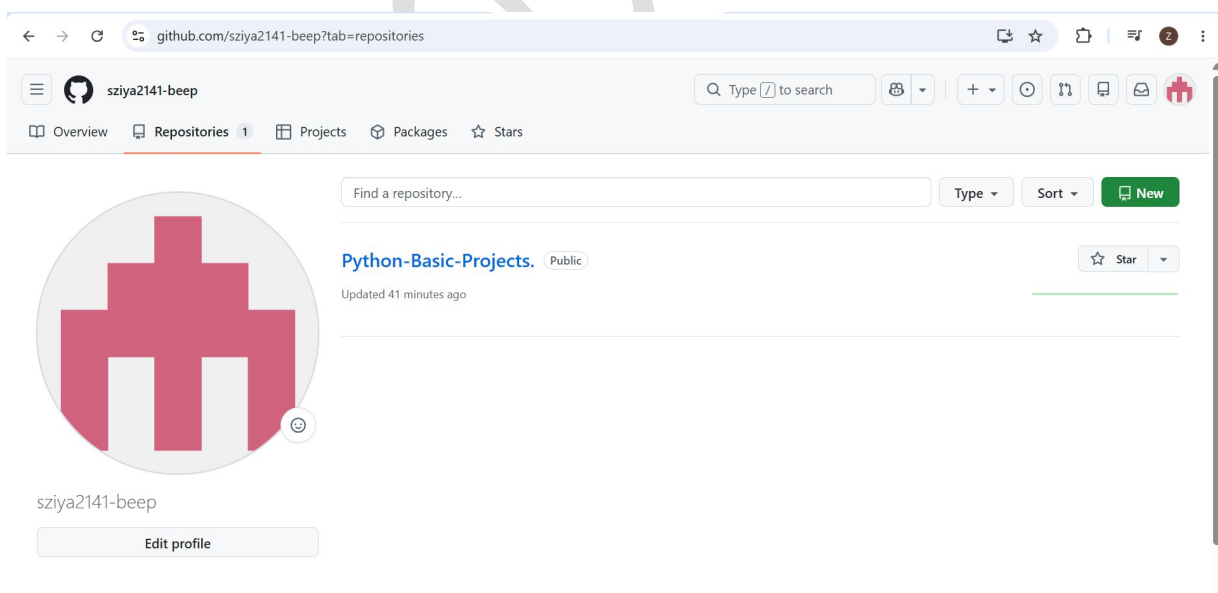
Domain Name : AIML Development

College Name : Zeal Polytechnic, pune

Branch Name : AIML

DATE: 01.07.2026, WEDNESDAY

Q1:



Q2:

```
Task 12.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 12.py (3.13)
File Edit Format Run Options Window Help

def analyze_string(s):
    print("\nString Length:", len(s))

    print("Reversed String:", s[::-1])

    vowels = "aeiouAEIOU"
    count = 0

    for ch in s:
        if ch in vowels:
            count += 1

    print("Number of vowels:", count)

    print("\nCharacter with Positive and Negative Index")

    for i in range(len(s)):
        print(f"{s[i]} -> Positive Index: {i}, Negative Index: {i-len(s)}")

user_string = input("Enter a string: ")

if user_string == "":
    print("Empty string entered.")
else:
    analyze_string(user_string)
```

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943]
Enter "help" below or click "Help" above for more information.

>>>

=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/
Enter a string: Ziya Shaikh

String Length: 11

Reversed String: hkiahS ayiZ

Number of vowels: 4

Character with Positive and Negative Index

Z -> Positive Index: 0, Negative Index: -11

i -> Positive Index: 1, Negative Index: -10

y -> Positive Index: 2, Negative Index: -9

a -> Positive Index: 3, Negative Index: -8

| -> Positive Index: 4, Negative Index: -7

S -> Positive Index: 5, Negative Index: -6

h -> Positive Index: 6, Negative Index: -5

a -> Positive Index: 7, Negative Index: -4

i -> Positive Index: 8, Negative Index: -3

k -> Positive Index: 9, Negative Index: -2

h -> Positive Index: 10, Negative Index: -1

>>>

Q3:

Task 12.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 12.py (3.13.5)

File Edit Format Run Options Window Help

```
# Function to calculate average
```

```
def calculate_average(numbers):
```

```
    return sum(numbers) / len(numbers)
```

```
# Main Program
```

```
try:
```

```
    n = int(input("How many numbers do you want to enter? "))
```

```
    if n <= 0:
```

```
        raise ValueError("Number of elements must be greater than 0.")
```

```
    numbers = []
```

```
    for i in range(n):
```

```
        num = float(input(f"Enter number {i + 1}: "))
```

```
        numbers.append(num)
```

```
    print("\nNumbers entered:", numbers)
```

```
    print("Largest Number:", max(numbers))
```

```
    print("Smallest Number:", min(numbers))
```

```
    print("Average:", calculate_average(numbers))
```

```
except ValueError as e:
```

```
    print("Error:", e)
```

```
except ZeroDivisionError:
```

```
    print("Cannot calculate average of an empty list.")
```

```
except Exception as e:
```

```
    print("Unexpected Error:", e)
```

```
|
```

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 12.py ===
How many numbers do you want to enter? 7
Enter number 1: 56
Enter number 2: 67
Enter number 3: 67
Enter number 4: 77
Enter number 5: 55
Enter number 6: 55
Enter number 7: 44

Numbers entered: [56.0, 67.0, 67.0, 77.0, 55.0, 55.0, 44.0]
Largest Number: 77.0
Smallest Number: 44.0
Average: 60.142857142857146
>> |
```

Q4:

Task 12.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 12.py

File Edit Format Run Options Window Help

Class to store student details

class Student:

def __init__(self, name, marks):

self.name = name

self.marks = marks

def display(self):

print("\nStudent Name :", self.name)

print("Marks :", self.marks)

print("Average Marks :", sum(self.marks) / len(self.marks))

Main Program

students = []

try:

n = int(input("Enter number of students: "))

for i in range(n):

print(f"\nEnter details of Student {i+1}")

name = input("Enter Name: ")

marks = []

for j in range(3):

mark = float(input(f"Enter Marks of Subject {j+1}: "))

marks.append(mark)

students.append(Student(name, marks))

print("\n— Student Details —")

for student in students:

student.display()

```
# Main Program
students = []

try:
    n = int(input("Enter number of students: "))

    for i in range(n):
        print(f"\nEnter details of Student {i+1}")

        name = input("Enter Name: ")
        marks = []

        for j in range(3):
            mark = float(input(f"Enter Marks of Subject {j+1}: "))
            marks.append(mark)

        students.append(Student(name, marks))

    print("\n— Student Details —")

    for student in students:
        student.display()

except ValueError:
    print("Invalid input! Please enter numeric values for marks.")

except Exception as e:
    print("Error:", e)
```

Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
Enter "help" below or click "Help" above for more information.

>>>

=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 12.py ===

Enter number of students: 2

Enter details of Student 1

Enter Name: Ziya Shaikh

Enter Marks of Subject 1: 88

Enter Marks of Subject 2: 99

Enter Marks of Subject 3: 88

Enter details of Student 2

Enter Name: Ayesha Mirza

Enter Marks of Subject 1: 88

Enter Marks of Subject 2: 77

Enter Marks of Subject 3: 66

— Student Details —

Student Name : Ziya Shaikh

Marks : [88.0, 99.0, 88.0]

Average Marks : 91.66666666666667

Student Name : Ayesha Mirza

Marks : [88.0, 77.0, 66.0]

Average Marks : 77.0

>>> |

Q5:

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 12.py ===
Enter number of students: 2

Enter student name: Ziya
Enter marks of subject 1: 99
Enter marks of subject 2: 88
Enter marks of subject 3: 99

Enter student name: ayesha
Enter marks of subject 1: 88
Enter marks of subject 2: 77
Enter marks of subject 3: 77

— Student Report —

Name: Ziya
Marks: [99.0, 88.0, 99.0]
Average: 95.33333333333333
Grade: A+

Name: ayesha
Marks: [88.0, 77.0, 77.0]
Average: 80.66666666666667
Grade: A
>>>
```

```

    else:
        return "Fail"

# Dictionary to store student data
students = {}

n = int(input("Enter number of students: "))

for i in range(n):
    name = input("\nEnter student name: ")

    marks = []
    total = 0

    for j in range(3):
        mark = float(input(f"Enter marks of subject {j+1}: "))
        marks.append(mark)
        total += mark

    average = total / 3
    grade = calculate_grade(average)

    students[name] = {
        "Marks": marks,
        "Average": average,
        "Grade": grade
    }

print("\n— Student Report —")

for name, details in students.items():
    print("\nName:", name)
    print("Marks:", details["Marks"])
    print("Average:", details["Average"])
    print("Grade:", details["Grade"])

```

```

    average = total / 3
    grade = calculate_grade(average)

    students[name] = {
        "Marks": marks,
        "Average": average,
        "Grade": grade
    }

print("\n— Student Report —")

for name, details in students.items():
    print("\nName:", name)
    print("Marks:", details["Marks"])
    print("Average:", details["Average"])
    print("Grade:", details["Grade"])

```

Q6:

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64-bit]
Enter "help" below or click "Help" above for more information.

>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Python.exe
Enter number of students: 2

Enter student name: ziya
Enter marks of subject 1: 88
Enter marks of subject 2: 99
Enter marks of subject 3: 88

Enter student name: Ayesha
Enter marks of subject 1: 45
Enter marks of subject 2: 55
Enter marks of subject 3: 60

— Student Report —

Name: ziya
Marks: [88.0, 99.0, 88.0]
Average: 91.66666666666667
Grade: A+

Name: Ayesha
Marks: [45.0, 55.0, 60.0]
Average: 53.333333333333336
Grade: D

>>>
```

Function to calculate grade

def calculate_grade(avg):

if avg >= 90:

return "A+"

elif avg >= 80:

return "A"

elif avg >= 70:

return "B"

elif avg >= 60:

return "C"

elif avg >= 50:

return "D"

else:

return "Fail"

Dictionary to store student data

students = {}

n = int(input("Enter number of students: "))

for i in range(n):

name = input("\nEnter student name: ")

marks = []

total = 0

for j in range(3):

mark = float(input(f"Enter marks of subject {j+1}: "))

marks.append(mark)

total += mark

average = total / 3

grade = calculate_grade(average)

students[name] = {

marks, grade

```
average = total / 3
grade = calculate_grade(average)
```

```
students[name] = {
    "Marks": marks,
    "Average": average,
    "Grade": grade
}
```

```
print("\n— Student Report —")
```

```
for name, details in students.items():
    print("\nName:", name)
    print("Marks:", details["Marks"])
    print("Average:", details["Average"])
    print("Grade:", details["Grade"])
```

Q7:

Task 12.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 12.py (3.13.5)

File Edit Format Run Options Window Help

Program to create a list of numbers and perform operations using lambda

try:

n = int(input("Enter the value of n: "))

if n <= 0:

raise ValueError("Please enter a positive number.")

numbers = list(range(1, n + 1))

print("\nOriginal List:", numbers)

Square of each number using lambda

square = list(map(lambda x: x ** 2, numbers))

print("Square List:", square)

Even numbers using lambda

even = list(filter(lambda x: x % 2 == 0, numbers))

print("Even Numbers:", even)

Sum of all numbers

print("Sum of Numbers:", sum(numbers))

except ValueError as e:

print("Error:", e)

except Exception as e:

print("Unexpected Error:", e)

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 12.py ===
Enter the value of n: 44

Original List: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28,
29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44]
Square List: [1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400, 441, 48
4, 529, 576, 625, 676, 729, 784, 841, 900, 961, 1024, 1089, 1156, 1225, 1296, 1369, 1444, 1521, 1600,
1681, 1764, 1849, 1936]
Even Numbers: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44]
Sum of Numbers: 990
>>> |
```

Q8:

Task 12.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 12.py (3.13.5)

File Edit Format Run Options Window Help

```
# Class for Student
class Student:

    def __init__(self, roll_no, name, marks):
        self.roll_no = roll_no
        self.name = name
        self.marks = marks

    def display(self):
        print("\nRoll No:", self.roll_no)
        print("Name:", self.name)
        print("Marks:", self.marks)
        print("Average:", sum(self.marks) / len(self.marks))

students = {}

n = int(input("Enter number of students: "))

for i in range(n):
    roll = int(input("\nEnter Roll No: "))
    name = input("Enter Name: ")

    marks = (
        float(input("Enter Marks 1: ")),
        float(input("Enter Marks 2: ")),
        float(input("Enter Marks 3: "))
    )

    students[roll] = Student(roll, name, marks)

print("\n— Student Records —")

for student in students.values():
    student.display()
```

Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on w
Enter "help" below or click "Help" above for more information.

>>>

=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 12.py ===

Enter number of students: 2

Enter Roll No: 49

Enter Name: Ziya

Enter Marks 1: 99

Enter Marks 2: 88

Enter Marks 3: 87

Enter Roll No: 46

Enter Name: Sona

Enter Marks 1: 45

Enter Marks 2: 65

Enter Marks 3: 66

— Student Records —

Roll No: 49

Name: Ziya

Marks: (99.0, 88.0, 87.0)

Average: 91.33333333333333

Roll No: 46

Name: Sona

Marks: (45.0, 65.0, 66.0)

Average: 58.666666666666664

>>> |

Q9:

Task 12.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 12.py (3.13.5)

File Edit Format Run Options Window Help

import math

try:

```
sentence = input("Enter a sentence: ")
```

```
if sentence.strip() == "":
```

```
    raise ValueError("Sentence cannot be empty.")
```

```
words = sentence.lower().split()
```

```
unique_words = set(words)
```

```
print("\nUnique Words:")
```

```
print(sorted(unique_words))
```

```
count = len(unique_words)
```

```
print("Number of Unique Words:", count)
```

```
print("Square of Count:", math.pow(count, 2))
```

```
except ValueError as e:
```

```
    print(e)
```

```
except Exception as e:
```

```
    print("Error:", e)
```

IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>

=== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 12.py ===

Enter a sentence: Ziya is a good girl

Unique Words:

['a', 'girl', 'good', 'is', 'ziya']

Number of Unique Words: 5

Square of Count: 25.0

>>>

Q10:

```
*IDLE Shell 3.13.5*
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
==== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/Task 12.py ==

===== SMART CALCULATOR & DATA MANAGER =====
1. Add Student
2. View Students
3. Search Student
4. Calculator
5. Exit
Enter Your Choice: 1
Enter Roll Number: 7
Enter Student Name: Ziya
Enter Marks of Subject 1: 99
Enter Marks of Subject 2: 88
Enter Marks of Subject 3: 78
Student added successfully!

===== SMART CALCULATOR & DATA MANAGER =====
1. Add Student
2. View Students
3. Search Student
4. Calculator
5. Exit
Enter Your Choice: 2

— Student Records —

Roll No : 7
Name : Ziya
Marks : [99.0, 88.0, 78.0]
Average : 88.33
```

```
*IDLE Shell 3.13.5*
File Edit Shell Debug Options Window Help
Name: Ziya
Marks : [99.0, 88.0, 78.0]
Average : 88.33

===== SMART CALCULATOR & DATA MANAGER =====
1. Add Student
2. View Students
3. Search Student
4. Calculator
5. Exit
Enter Your Choice: 3
Enter Roll Number to Search: 7

Student Found
Name: Ziya
Marks: [99.0, 88.0, 78.0]
Average: 88.33

===== SMART CALCULATOR & DATA MANAGER =====
1. Add Student
2. View Students
3. Search Student
4. Calculator
5. Exit
Enter Your Choice: 4
Enter First Number: 77
Enter Second Number: 77
Enter Operation (+,-,*,./): *
Answer = 5929.0

===== SMART CALCULATOR & DATA MANAGER =====
1. Add Student
2. View Students
3. Search Student
4. Calculator
5. Exit
Enter Your Choice: |
```